

MA30091 2024/25 - Coursework 2

23538

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Gen AI Statement: I have used generative AI sparingly to reinforce my confidence and checking for spelling mistakes. All code and findings are my own.

Load in libraries to use

```
library(dplyr)
library(ggplot2)
library(readr)
library(tidyr)
library(MASS)
library(GGally)
library(forcats)
library(naniar)
library(mice)
library(knitr)
```

Analysing NHANES 2017/18 survey data

```
load(file="NHANES2017.rda")
```

Task 1

Perform an exploratory analysis of the amount and type(s) of missingness in the data. Write a summary of your findings and suggest reasons why data may be missing. [10 marks]

First I use the summary function to check the dataset for missing values and for any issues with the data.

```
data = nhanes2017
summary(data)
```

```
##           id           gender           age
##  Min.      : 93711   Male :1575   Min.      :21.00
##  1st Qu.: 96018   Female:1768  1st Qu.:30.00
##  Median : 98302                      Median :40.00
##  Mean    : 98317                      Mean    :40.32
##  3rd Qu.:100626                      3rd Qu.:51.00
##  Max.    :102956                      Max.    :59.00
##
```

```

##                                     ethnic
## Mexican American                   : 499
## Other Hispanic                     : 313
## Non-Hispanic White                 :1000
## Non-Hispanic Black                 : 765
## Non-Hispanic Asian                 : 575
## Other Race - Including Multi-Racial: 191
##
##                                     educ
## Less than 9th grade                 : 216
## 9-11th grade (Includes 12th grade with no diploma): 361
## High school graduate/GED or equivalent : 765
## Some college or AA degree           :1132
## College graduate or above           : 866
## Refused                             :   1
## Don't Know                          :   2
##
##          marital          income          weights          height
## Married          :1596 low          : 588 Min.      : 4363 Min.      :138.3
## Widowed          :   53 high         :2425 1st Qu.: 19639 1st Qu.:159.7
## Divorced         : 302 Refused      :   69 Median   : 30245 Median :166.7
## Separated        : 130 Don't know:   88 Mean      : 49395 Mean   :167.2
## Never married    : 828 NA's         : 173 3rd Qu.: 52477 3rd Qu.:174.6
## Living with partner: 432 Max.      :433085 Max.      :195.8
## Refused          :   2 NA's         :199
##
##          bmi          waist          glycohemo          cholesterol
## Min.      :14.80 Min.      : 57.90 Min.      : 4.100 Min.      : 27.0
## 1st Qu.:24.60 1st Qu.: 86.60 1st Qu.: 5.200 1st Qu.: 90.0
## Median :28.80 Median : 97.50 Median : 5.500 Median :110.0
## Mean      :30.11 Mean      : 99.62 Mean      : 5.678 Mean      :113.2
## 3rd Qu.:34.00 3rd Qu.:110.20 3rd Qu.: 5.800 3rd Qu.:133.0
## Max.      :86.20 Max.      :169.50 Max.      :16.200 Max.      :354.0
## NA's      :203 NA's      :326 NA's      :329 NA's      :1918
##
## triglyceride          vwa          mwa          wb          vrvav
## Min.      : 10.0 Yes          : 950 Yes          :1527 Yes: 895 Yes:1031
## 1st Qu.: 58.0 No           :2390 No           :1814 No :2448 No :2312
## Median : 89.0 Don't know:   3 Don't know:   2
## Mean      :115.5
## 3rd Qu.:137.0
## Max.      :2684.0
## NA's      :1896
##
## mrav          msav          sm100
## Yes:1446 Min.      :   2.0 Yes:1261
## No :1897 1st Qu.:180.0 No :2082
##
##          Median : 300.0
##          Mean    : 361.5
##          3rd Qu.: 480.0
##          Max.    :9999.0
##          NA's    :8

```

There is a lot of missing data, encoded as NA, in the examination and laboratory data. Namely, the most amounts of missing data comes from cholesterol and triglyceride. Some of the missing data is encoded as “Don’t know” or “Refusal”, seen in the demographic and questionnaire data: the refusals are missing not at random (MNAR) as they likely do not want to share personal data. There is a lot of structural missing data, since there are known reasons to why this information is not being shared. As for the examination and

laboratory data, there could be missingness due to errors in the processing and/or measurements not being taken. For this analysis of missingness, I keep all missing values as they have been encoded as the reasons for this are likely to differ. For example, the NA values for income are likely people who are unemployed, but 'Don't Know' or 'Refusal' are because they do not wish to disclose their earnings but are employed.

There is a maximum value for triglyceride of 2684- this is extremely unlikely but I cannot confirm whether this is an error or not so will be left as it is for analysis.

There is 1440 minutes in a day so this should be the maximum value for msav. Suggesting that there is also an issue with this variable.

```
msav_subset <- data[!is.na(data$msav) & data$msav >= 1440, ]
msav_subset$msav
```

```
## [1] 9999 9999 9999 9999 9999 9999 9999 9999 9999 9999 9999 9999
```

All 12 participants with a msav value of 9999 are the only ones with a value higher than 1440- this has probably been inputted incorrectly and should be recoded to take value NA

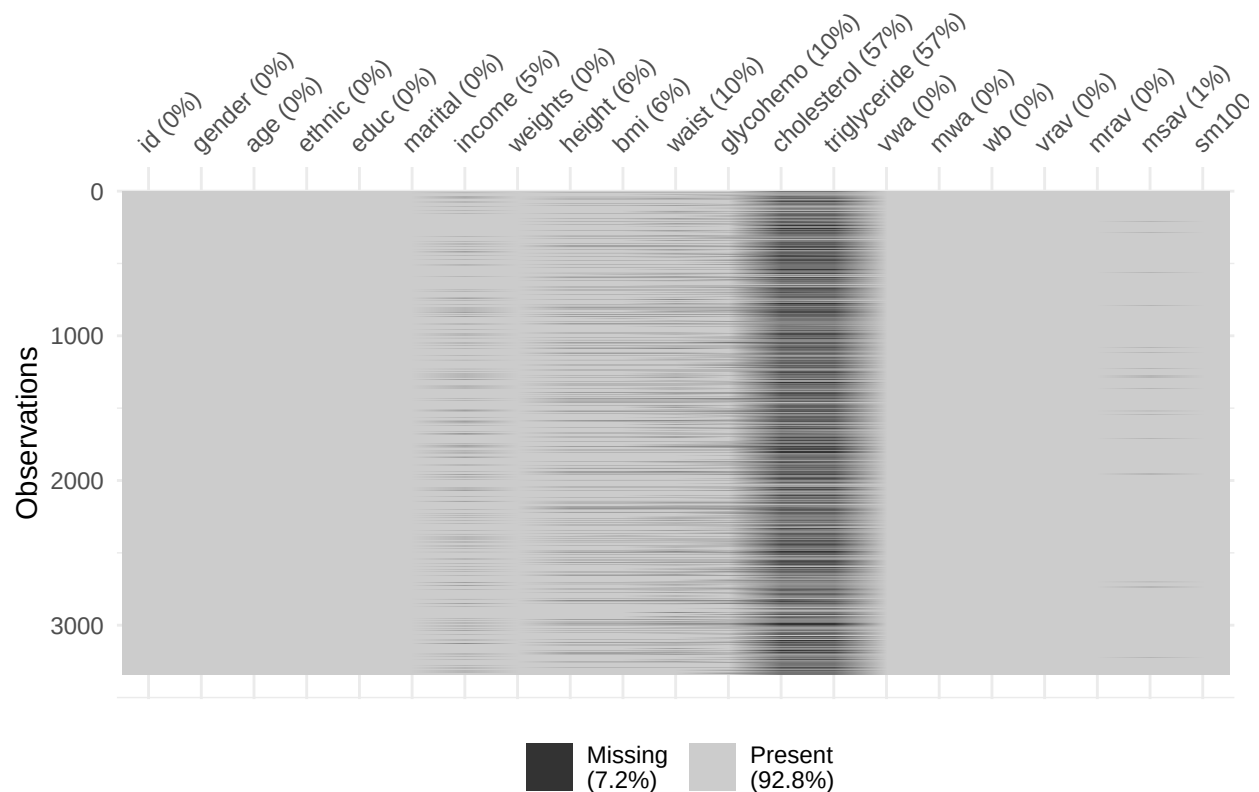
```
data$msav[data$msav>=1440]=NA
```

Now let's investigate the NA values: To analyse missingness let's look at the proportion of missing values:

```
apply(is.na(data),2,mean)
```

```
##          id          gender          age          ethnic          educ          marital
## 0.00000000 0.00000000 0.00000000 0.00000000 0.00000000 0.00000000
##      income      weights      height          bmi          waist      glycohem
## 0.05174993 0.00000000 0.05952737 0.06072390 0.09751720 0.09841460
## cholesterol triglyceride          vwa          mwa          wb          vrav
## 0.57373617 0.56715525 0.00000000 0.00000000 0.00000000 0.00000000
##          mrav          msav          sm100
## 0.00000000 0.00598265 0.00000000
```

```
vis_miss(data)
```



Missingness doesn't seem to be less or more severe in any areas of data points, there is no obvious pattern. This does visually show how much of cholesterol and triglyceride is missing compared to other variables.

```
mean(apply(is.na(data), 1, max))
```

```
## [1] 0.6084355
```

There is missing data for 60% of observations. Missing values occur for, income(~5%), height (~6%), bmi(~6%), waist(~10%), glycohem(~10%), cholesterol(~57%) and triglyceride(~57%).

The variables with highest percent of missing data is cholesterol and triglyceride- let's explore what type of missing data this is. I need to check to see if there is any correlation between the missing data and any variables.

I need to check if the participants with missing values for cholesterol also have it for triglyceride. Although there are lower number of missing values for glycohem I will still see if there are any overlaps:

```
sum(is.na(data$cholesterol) & is.na(data$triglyceride))
```

```
## [1] 1896
```

```
(sum(is.na(data$cholesterol) & is.na(data$triglyceride)) / nrow(data)) * 100
```

```
## [1] 56.71552
```

```
sum(!is.na(data$cholesterol) & is.na(data$glycohemmo))
```

```
## [1] 2
```

```
(sum(is.na(data$cholesterol) & is.na(data$glycohemmo)) / nrow(data)) * 100
```

```
## [1] 9.781633
```

```
sum(!is.na(data$triglyceride) & is.na(data$glycohemmo))
```

```
## [1] 2
```

```
(sum(is.na(data$triglyceride) & is.na(data$glycohemmo)) / nrow(data)) * 100
```

```
## [1] 9.781633
```

There are 1896 observations with both cholesterol and triglyceride missing- this accounts for 56% of the data. It is likely that a blood test for both could not be examined properly or a blood test was never conducted. This means that missing values for cholesterol and triglyceride are unlikely to be because of randomness- there is a reason for this. However, the NA values are very unlikely to be due to the actual value that is missing and therefore this data is MNAR- the reasons for this could be due to job or age affecting whether a person has time for a blood test. For the values that are only missing one of the laboratory data- this could be due to errors in testing the blood or wrong tests being ordered.

There are only 2 observations which are missing data for glycohemmo but have values for triglyceride and cholesterol- so the majority either have values for all or are missing all three. This backs up my theory before for missing data, glycohemmo can be found through a different blood test to the other two- so it seems that participants missing all three, likely didn't attend a blood test.

Other values that have missing 'NA' values are: income, height, bmi, waist. Let's investigate if there are any reasons for these: For income, I would suggest that maybe these are missing due to different levels of earning- perhaps these are people who are unemployed and do not have a value, or people who have refused to answer but was encoded differently.

```
income_missing <- is.na(data$income)
cor(data$age, income_missing, use="pairwise")
```

```
## [1] -0.01678573
```

```
prop.table(xtabs(~income_missing + data$educ))
```

```
##           data$educ
## income_missing Less than 9th grade
##           FALSE      0.0616212982
##           TRUE       0.0029913252
##           data$educ
## income_missing 9-11th grade (Includes 12th grade with no diploma)
##           FALSE      0.1026024529
##           TRUE       0.0053843853
##           data$educ
```

```
## income_missing High school graduate/GED or equivalent Some college or AA degree
##          FALSE                                0.2186658690          0.3206700568
##          TRUE                                 0.0101705055          0.0179479509
##          data$educ
## income_missing College graduate or above      Refused    Don't Know
##          FALSE          0.2437930003 0.0002991325 0.0005982650
##          TRUE          0.0152557583 0.0000000000 0.0000000000
```

There is a slight increase in proportion of missing data for education levels for high school graduates and above- suggesting that this might be due to not wanting to answer rather than no earnings. There is also no correlation between missing income data and age- this is expected as all participants are between working ages so I would expect a low number of unemployed participants in this setting. There is probably some slight MNAR but hard to know for sure. These missing values seem to be MNAR but there could be slight missing at random with correlation between variables such as income and education levels.

Height, BMI and Waist: Let's check for any overlap between these variables:

```
sum(is.na(data$height) & is.na(data$bmi))
```

```
## [1] 199
```

```
(sum(is.na(data$height) & is.na(data$bmi)) / nrow(data)) * 100
```

```
## [1] 5.952737
```

```
sum(!is.na(data$height) & is.na(data$waist))
```

```
## [1] 129
```

```
(sum(is.na(data$height) & is.na(data$waist)) / nrow(data)) * 100
```

```
## [1] 5.892911
```

```
sum(!is.na(data$bmi) & is.na(data$waist))
```

```
## [1] 128
```

```
(sum(is.na(data$bmi) & is.na(data$waist)) / nrow(data)) * 100
```

```
## [1] 5.922824
```

All proportions of missing data are similar to values I observe when looking at proportion of missing data for each variable. This suggests that participants with missing examination data, have it missing for all examination data. Likely, this is due to social desirability so these are probably the lower and higher values I would expect. Waist has a higher percent of missing data- this is likely due to participants not feeling comfortable with this measurement being taken. So the missing examination data is MNAR- it is likely that the reason for missingness is related to the value that is missing, leading to selection bias.

There are also 'missing' values encoded as "Don't Know" and "Refused". I do not want to change these to be "NA" as they are likely to be MNAR and the reasons are more obvious than "NA" values. Therefore I will keep these variables as they are, they are still meaningful. For example, in the variable "Income" there are 88 "Don't Knows" and 69 "Refusals"- it is likely that there is some overlap and that people answer don't know as they don't want to refuse but are embarrassed by their answer. These refusals to answer are likely to be from people at the lower end of "low" income and higher end of "high" income.

Task 2

Using the weights in the dataset estimate the national percentage of individuals who smoke 100 cigarettes in life for the population in 2017/18 in USA. Estimate the same percentage as per ethnicity, education category and income. Compare these estimates with the one you may obtain from the sample without using the weights in the dataset. Comment on your findings. [10 marks]

I first find the national percentage using the “weights” variable which adjusts for selection and non-response bias:

```
#Create an indicator variable for whether a person smokes or not
data$n_sm100[data$sm100=='Yes']=1
data$n_sm100[data$sm100=='No']=0
weighted_smk= data %>% summarise(sum(n_sm100 * weights) / sum(weights))
colnames(weighted_smk) <- "Weighted Smoking Percentage" #Changing column name
weighted_smk
```

```
##   Weighted Smoking Percentage
## 1                      0.4060647
```

40.6% of the USA in 2017/18 smoke 100 cigarettes in the entire life.

Now I calculate the estimates per ethnicity, education and income using the same indicator variable I previously created.

```
ethnic_weight <- data %>% group_by(ethnic) %>%
  summarise((weighted_smk=sum(n_sm100*weights)/sum(weights))*100)
ethnic_weight <- setNames(ethnic_weight, c("Ethnicity", "Weighted Smoking %"))
ethnic_weight
```

```
## # A tibble: 6 x 2
##   Ethnicity                'Weighted Smoking %'
##   <fct>                  <dbl>
## 1 Mexican American        34.6
## 2 Other Hispanic          29.4
## 3 Non-Hispanic White      46.0
## 4 Non-Hispanic Black      32.3
## 5 Non-Hispanic Asian      17.1
## 6 Other Race - Including Multi-Racial  58.8
```

```
educ_weight <- data %>% group_by(educ) %>%
  summarise((weighted_smk=sum(n_sm100*weights)/sum(weights))*100)
educ_weight <- setNames(educ_weight, c("Education Level", "Weighted Smoking %"))
educ_weight
```

```
## # A tibble: 7 x 2
##   'Education Level'                'Weighted Smoking %'
##   <fct>                  <dbl>
## 1 Less than 9th grade          37.8
## 2 9-11th grade (Includes 12th grade with no diploma)  56.4
## 3 High school graduate/GED or equivalent  51.9
## 4 Some college or AA degree      43.0
## 5 College graduate or above      25.8
## 6 Refused                        0
## 7 Don't Know                    0
```

```
inc_weight <- data %>% group_by(income) %>%
  summarise((weighted_smk=sum(n_sm100*weights)/sum(weights))*100)
inc_weight <- setNames(inc_weight, c("Income", "Weighted Smoking %"))
inc_weight
```

```
## # A tibble: 5 x 2
##   Income      'Weighted Smoking %'
##   <fct>          <dbl>
## 1 low           52.0
## 2 high          38.9
## 3 Refused       51.8
## 4 Don't know    24.1
## 5 <NA>          40.5
```

Then find unweighted values and create a table comparing the two:

```
ethnic_weight <- data %>% group_by(ethnic) %>%
  summarise((weighted_smk=sum(n_sm100*weights)/sum(weights))*100, unweighted_smk=mean(n_sm100)*100)
ethnic_weight <- setNames(ethnic_weight, c("Ethnicity", "Weighted Smoking %", "Unweighted Smoking %"))
ethnic_weight
```

```
## # A tibble: 6 x 3
##   Ethnicity      'Weighted Smoking %' 'Unweighted Smoking %'
##   <fct>          <dbl>          <dbl>
## 1 Mexican American    34.6          33.3
## 2 Other Hispanic      29.4          28.8
## 3 Non-Hispanic White  46.0          53
## 4 Non-Hispanic Black  32.3          35.0
## 5 Non-Hispanic Asian  17.1          18.4
## 6 Other Race - Including Multi-Raci~ 58.8          52.9
```

```
educ_weight <- data %>% group_by(educ) %>%
  summarise((weighted_smk=sum(n_sm100*weights)/sum(weights))*100, unweighted_smk=mean(n_sm100)*100)
educ_weight <- setNames(educ_weight, c("Education Level", "Weighted Smoking %", "Unweighted Smoking %"))
educ_weight
```

```
## # A tibble: 7 x 3
##   'Education Level'      'Weighted Smoking %' 'Unweighted Smoking %'
##   <fct>          <dbl>          <dbl>
## 1 Less than 9th grade    37.8          33.8
## 2 9-11th grade (Includes 12th grade~ 56.4          52.4
## 3 High school graduate/GED or equiv~ 51.9          47.1
## 4 Some college or AA degree 43.0          41.3
## 5 College graduate or above 25.8          19.7
## 6 Refused              0              0
## 7 Don't Know            0              0
```

```
inc_weight <- data %>% group_by(income) %>%
  summarise((weighted_smk=sum(n_sm100*weights)/sum(weights))*100, unweighted_smk=mean(n_sm100)*100)
inc_weight <- setNames(inc_weight, c("Income", "Weighted Smoking %", "Unweighted Smoking %"))
inc_weight
```



```
## # A tibble: 5 x 3
##   Income      'Weighted Smoking %' 'Unweighted Smoking %'
##   <fct>          <dbl>          <dbl>
## 1 low           52.0           49.3
## 2 high          38.9           35.6
## 3 Refused       51.8           39.1
## 4 Don't know    24.1           23.9
## 5 <NA>          40.5           34.7
```

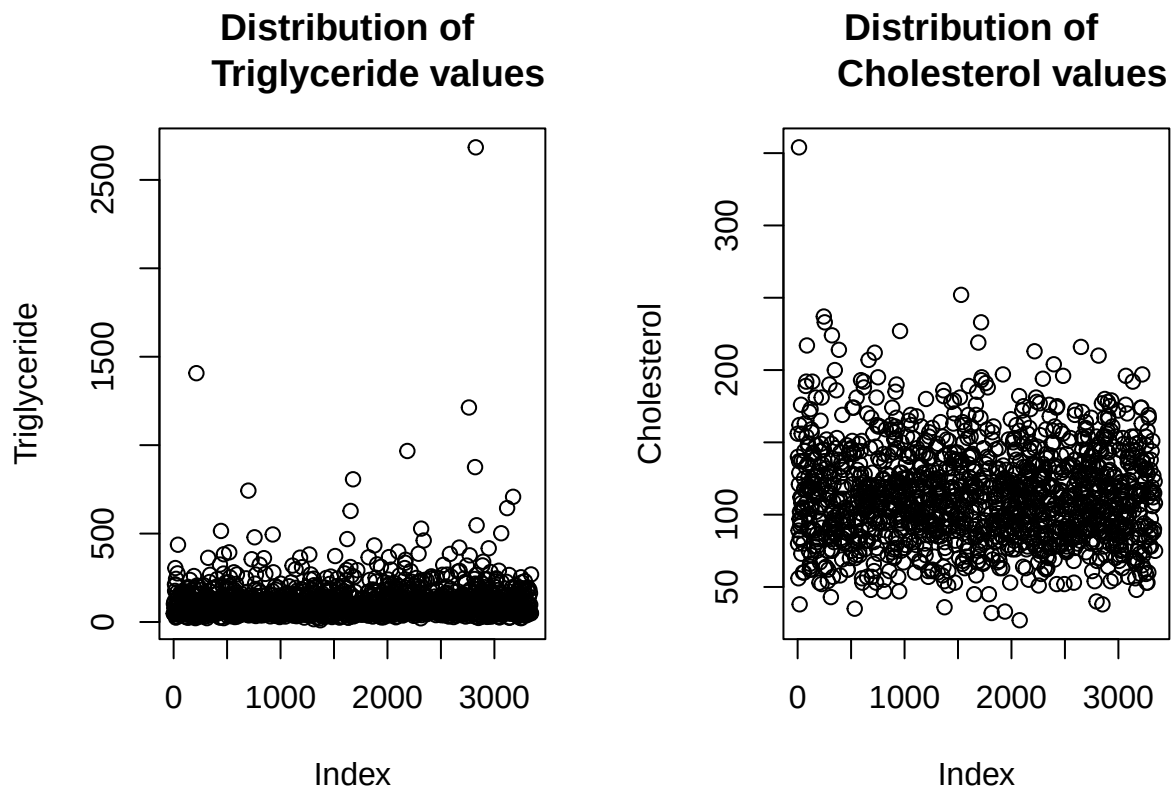
The unweighted smoking percent of people who smoke at least 100 cigarettes in their entire life is quite similar to the weighted percentage. The highest percentage I found for the weighted calculation for ethnicity is in the “Other Race” category, whereas “Non-Hispanic White” is approximately 0.2% higher. The ethnicity which smoke the least are Non-Hispanic Asians with approximately 17% weighted percentage and 18% unweighted percentage. For education level, ignoring the 0% average that refused or didn’t answer, the highest weighted percentage is from participants with 9-11th grade education and this is the same for unweighted percentage. The education level achieved which has the lowest number of people smoking at least 100 cigarettes in their lifetime for both weighted and unweighted are those who are college graduates or above. For income levels, approximately 41% (35% unweighted) of people who did not answer, encoded as “NA” smoked at least 100 cigarettes in their lifetime. The percentage of people smoking 100 cigarettes in their lifetime are highest for those with low incomes, approximately 52% (and 49% unweighted).

Task 3

Fit an appropriate model to examine the risk of cardiovascular disease in the participants in the survey. Write a summary which details your approach and results. Your summary should also clearly motivate your choice of model and discuss the validity of your model assumptions. [15 marks]

By simply plotting the variables that are indicators for cardiovascular disease I observe that there is no skewness in the data so I fitted a quasi binomial generalised linear model. I chose this since there is a lot of variation within the model, and the outcome is binary- a person is either at risk of cardiovascular disease or is not. The variables that are indicators for cardiovascular disease that are used are cholesterol or triglyceride, the plots below show that the values in the data are unskewed.

```
par(mfrow=c(1,2))
plot(data$triglyceride, ylab="Triglyceride", main="Distribution of
      Triglyceride values")
plot(data$cholesterol, ylab="Cholesterol", main="Distribution of
      Cholesterol values")
```



One thing worth mentioning here, is the highest value of triglyceride which I mentioned at the start of this report that may skew results slightly but there is not enough reasoning to exclude this data point.

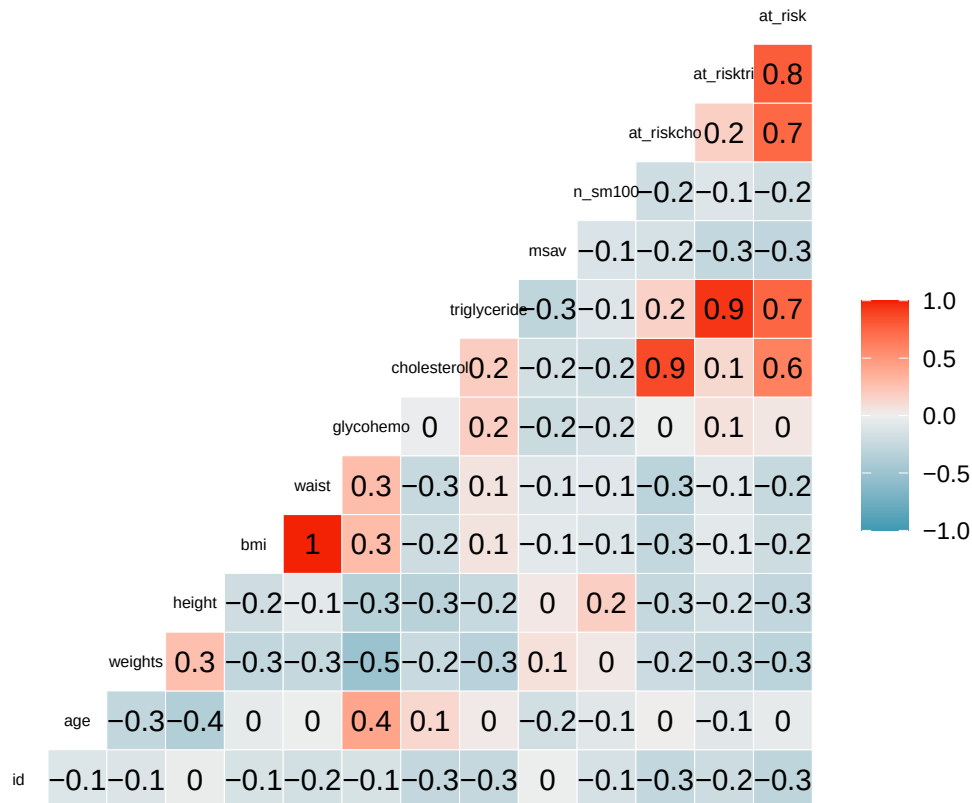
I create a variable which has a score for how at risk a participant is. If they are at risk based on their laboratory data- each 'at risk' score counts for 1. If a participant has a NA value for one cholesterol or triglyceride but a value for the other then the risk is calculated using the one value there is.

```
data$at_riskcho <- ifelse(data$cholesterol > 160, 1, 0)
data$at_risktri <- ifelse(data$triglyceride > 200, 1, 0)
data$at_risk <- data$at_riskcho + data$at_risktri
data$at_risk <- ifelse(data$at_riskcho == 1 | data$at_risktri == 1, 1,
  ifelse(data$at_riskcho == 0 & data$at_risktri == 0, 0, NA))
#Encodes 1 if at least one of the variables shows risk or 0
#if neither indicate the participant to be at risk of cardiovascular disease
```

First check if there is any collinearity that would result in us omitting some variables from this model

```
## Create a new data frame which contains only numeric variables
numerical_vars <- data[, sapply(data, is.numeric)]

## Calculate the correlation between each pair of numerical variables in the dataset
numerical_vars %>%
  cor(use = "complete.obs") %>%
  ggcorr(label = TRUE, size = 2, label_round = 0.8)
```



Waist and BMI are perfectly correlated so remove one of these variables. BMI includes the height measurement in its calculation I remove waist and height from the model but include BMI. vwa and mwa measures activity level and can be combined into one variable so I create a new variable which accounts for both, this is the same for vrv and mrv which measures intensity so I also create a new intensity variable.

```
data$activity <- case_when(
  data$vwa == "Yes" ~ "vigorous",
  data$mwa == "Yes" ~ "moderate",
  TRUE ~ "low")
data$intensity <- case_when(
  data$vrav == "Yes" ~ "vigorous",
  data$mrav == "Yes" ~ "moderate",
  TRUE ~ "low")
```

I chose not to use weights in the linear model as I included a lot of variables in the model and it aids convergence of the quasi binomial model.

```
modell1 <- glm(data=data,
  at_risk~gender+age+income+marital+ethnic+bmi+activity+wb+intensity+msav+sm100,
  family=quasibinomial)
summary(modell1)
```

```
##
## Call:
## glm(formula = at_risk ~ gender + age + income + marital + ethnic +
##      bmi + activity + wb + intensity + msav + sm100, family = quasibinomial,
```

```

##      data = data)
##
## Coefficients:
##
##              Estimate Std. Error t value
## (Intercept)      -2.6367656   0.5702235  -4.624
## genderFemale      -0.6120071   0.1559041  -3.926
## age                0.0181201   0.0074307   2.439
## incomehigh        -0.1811200   0.1971419  -0.919
## incomeRefused      0.4081606   0.5212078   0.783
## incomeDon't know  -0.9086074   0.6421193  -1.415
## maritalWidowed     0.0999635   0.5323372   0.188
## maritalDivorced    -0.0575661   0.2653420  -0.217
## maritalSeparated    0.6348547   0.3463995   1.833
## maritalNever married -0.2260654   0.2172786  -1.040
## maritalLiving with partner -0.1222235   0.2485782  -0.492
## ethnicOther Hispanic -0.0873628   0.2929788  -0.298
## ethnicNon-Hispanic White -0.4056012   0.2267546  -1.789
## ethnicNon-Hispanic Black -0.7860191   0.2619736  -3.000
## ethnicNon-Hispanic Asian  0.6273179   0.2561009   2.449
## ethnicOther Race - Including Multi-Racial -0.3042892   0.3313173  -0.918
## bmi                0.0299297   0.0092674   3.230
## activitymoderate    0.3771183   0.1959241   1.925
## activityvigorous    0.4850619   0.1857715   2.611
## wbNo               0.1884139   0.1727195   1.091
## intensitymoderate  -0.1575474   0.1878561  -0.839
## intensityvigorous  -0.3718239   0.1846715  -2.013
## msav               0.0002737   0.0003761   0.728
## sm100No            -0.3316082   0.1552544  -2.136
##
##              Pr(>|t|)
## (Intercept)      4.12e-06 ***
## genderFemale      9.09e-05 ***
## age                0.01487 *
## incomehigh        0.35840
## incomeRefused     0.43370
## incomeDon't know  0.15730
## maritalWidowed    0.85108
## maritalDivorced   0.82828
## maritalSeparated  0.06706 .
## maritalNever married 0.29832
## maritalLiving with partner 0.62302
## ethnicOther Hispanic 0.76561
## ethnicNon-Hispanic White 0.07388 .
## ethnicNon-Hispanic Black 0.00275 **
## ethnicNon-Hispanic Asian 0.01443 *
## ethnicOther Race - Including Multi-Racial 0.35856
## bmi               0.00127 **
## activitymoderate  0.05446 .
## activityvigorous  0.00913 **
## wbNo             0.27553
## intensitymoderate 0.40181
## intensityvigorous 0.04427 *
## msav             0.46690
## sm100No          0.03287 *
## ---

```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for quasibinomial family taken to be 0.9971952)
##
## Null deviance: 1295.5  on 1368  degrees of freedom
## Residual deviance: 1200.3  on 1345  degrees of freedom
## (1974 observations deleted due to missingness)
## AIC: NA
##
## Number of Fisher Scoring iterations: 5
```

There are a few variables that are not significant in this model, I start by removing the variables with high p-values: income, msav

```
modell1a <- glm(data=data,
               at_risk~gender+age+marital+ethnic+bmi+activity+wb+intensity+sm100,
               family=quasibinomial)
summary(modell1a)
```

```
##
## Call:
## glm(formula = at_risk ~ gender + age + marital + ethnic + bmi +
##       activity + wb + intensity + sm100, family = quasibinomial,
##       data = data)
##
## Coefficients:
##
## Estimate Std. Error t value Pr(>|t|)
## (Intercept) -2.811632 0.528017 -5.325 1.17e-07
## genderFemale -0.580459 0.152636 -3.803 0.000149
## age 0.020761 0.007300 2.844 0.004520
## maritalWidowed 0.161126 0.528091 0.305 0.760327
## maritalDivorced -0.033086 0.257835 -0.128 0.897912
## maritalSeparated 0.566508 0.342771 1.653 0.098610
## maritalNever married -0.151900 0.209495 -0.725 0.468527
## maritalLiving with partner -0.080494 0.243094 -0.331 0.740601
## ethnicOther Hispanic -0.140862 0.286712 -0.491 0.623290
## ethnicNon-Hispanic White -0.322898 0.217414 -1.485 0.137721
## ethnicNon-Hispanic Black -0.703769 0.250237 -2.812 0.004985
## ethnicNon-Hispanic Asian 0.595625 0.245167 2.429 0.015246
## ethnicOther Race - Including Multi-Racial -0.287300 0.324619 -0.885 0.376287
## bmi 0.029867 0.008992 3.321 0.000918
## activitymoderate 0.253693 0.190034 1.335 0.182097
## activityvigorous 0.372132 0.176693 2.106 0.035373
## wbNo 0.162710 0.167475 0.972 0.331443
## intensitymoderate -0.155923 0.182883 -0.853 0.394037
## intensityvigorous -0.376280 0.180261 -2.087 0.037030
## sm100No -0.305099 0.152877 -1.996 0.046158
##
## (Intercept) ***
## genderFemale ***
## age **
## maritalWidowed
## maritalDivorced
```

```
## maritalSeparated .
## maritalNever married
## maritalLiving with partner
## ethnicOther Hispanic
## ethnicNon-Hispanic White
## ethnicNon-Hispanic Black **
## ethnicNon-Hispanic Asian *
## ethnicOther Race - Including Multi-Racial
## bmi ***
## activitymoderate
## activityvigorous *
## wbNo
## intensitymoderate
## intensityvigorous *
## sm100No *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for quasibinomial family taken to be 1.005958)
##
## Null deviance: 1342.0 on 1424 degrees of freedom
## Residual deviance: 1256.8 on 1405 degrees of freedom
## (1918 observations deleted due to missingness)
## AIC: NA
##
## Number of Fisher Scoring iterations: 5
```

Next, should also remove marital, wb and sm100 here:

```
model1b <- glm(data=data,
               at_risk~gender+age+ethnic+bmi+activity+intensity, family=quasibinomial)
drop1(model1b)
```

```
## Single term deletions
##
## Model:
## at_risk ~ gender + age + ethnic + bmi + activity + intensity
##           Df Deviance
## <none>      1265.4
## gender      1  1281.9
## age         1  1278.7
## ethnic      5  1294.2
## bmi         1  1276.5
## activity    2  1270.8
## intensity   2  1271.5
```

```
summary(model1b)
```

```
##
## Call:
## glm(formula = at_risk ~ gender + age + ethnic + bmi + activity +
##       intensity, family = quasibinomial, data = data)
```

```
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -3.001097   0.461131  -6.508 1.05e-10
## genderFemale   -0.600792   0.148566  -4.044 5.54e-05
## age            0.023608   0.006540   3.610 0.000317
## ethnicOther Hispanic    -0.157642   0.283973  -0.555 0.578894
## ethnicNon-Hispanic White -0.265563   0.212868  -1.248 0.212403
## ethnicNon-Hispanic Black -0.730753   0.245073  -2.982 0.002915
## ethnicNon-Hispanic Asian  0.548888   0.237810   2.308 0.021138
## ethnicOther Race - Including Multi-Racial -0.248973   0.320276  -0.777 0.437071
## bmi            0.030264   0.008933   3.388 0.000723
## activitymoderate  0.257152   0.188213   1.366 0.172068
## activityvigorous  0.394895   0.174337   2.265 0.023656
## intensitymoderate -0.172187   0.180410  -0.954 0.340035
## intensityvigorous -0.432339   0.176715  -2.447 0.014545
##
## (Intercept)      ***
## genderFemale      ***
## age               ***
## ethnicOther Hispanic
## ethnicNon-Hispanic White
## ethnicNon-Hispanic Black      **
## ethnicNon-Hispanic Asian      *
## ethnicOther Race - Including Multi-Racial
## bmi               ***
## activitymoderate
## activityvigorous      *
## intensitymoderate
## intensityvigorous      *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for quasibinomial family taken to be 0.9971077)
##
## Null deviance: 1342.0 on 1424 degrees of freedom
## Residual deviance: 1265.4 on 1412 degrees of freedom
## (1918 observations deleted due to missingness)
## AIC: NA
##
## Number of Fisher Scoring iterations: 4
```

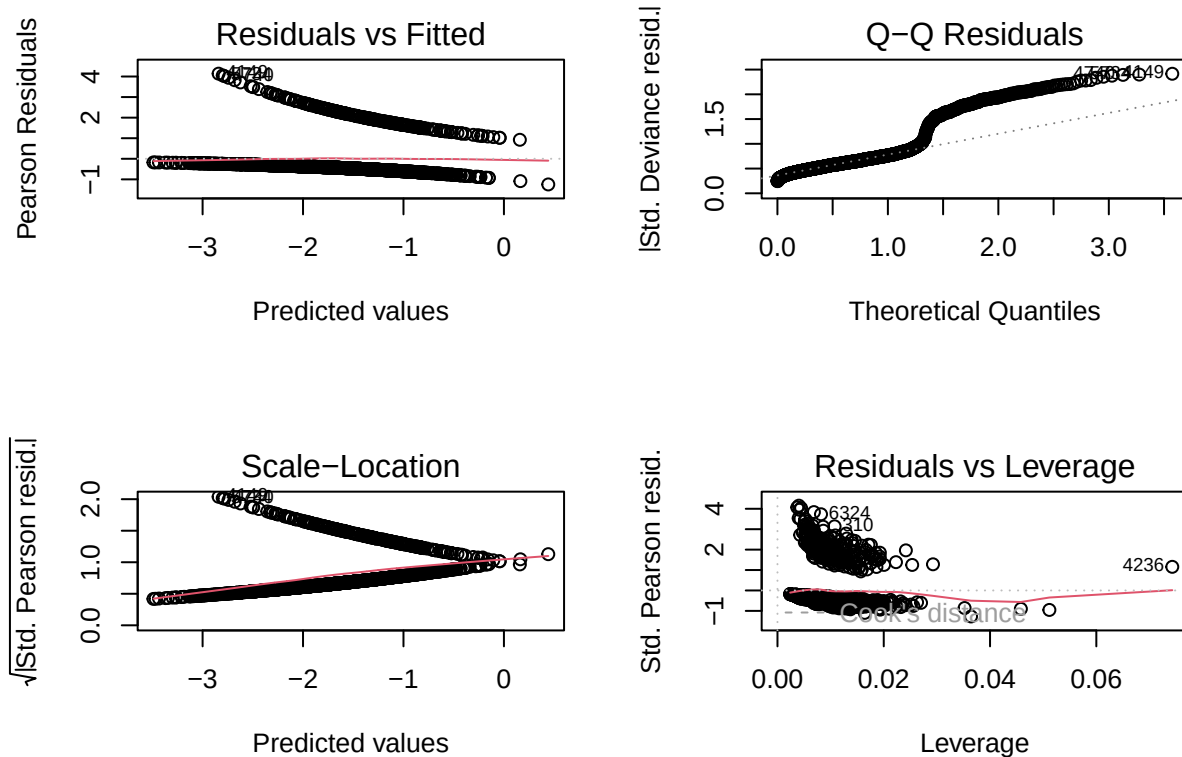
Ethnicity has higher deviance and some of the factors are not significant but overall it is, this could be due to varying number of observations:

```
summary(data$ethnic)
```

```
##           Mexican American           Other Hispanic
##                499                313
## Non-Hispanic White Non-Hispanic Black
##                1000                765
## Non-Hispanic Asian Other Race - Including Multi-Racial
##                575                191
```

This shows varying number of observations which can explain differences in deviance but I should not remove this variable from the model. Let's check the diagnostic plots for this model:

```
par(mfrow = c(2, 2))
plot(model1b)
```



There are some obvious issues with the diagnostic plots that I aim to fix. I shall try adding interaction terms to the model, using knowledge of the variables and how they behave I choose to explore only a few interactions:

```
model1c <- glm(data=data,
               at_risk~gender+age+ethnic+bmi+activity+intensity+age*bmi
               +age*activity+age*intensity+gender*bmi+activity*intensity,
               family=quasibinomial)
summary(model1c)
```

```
##
## Call:
## glm(formula = at_risk ~ gender + age + ethnic + bmi + activity +
##      intensity + age * bmi + age * activity + age * intensity +
##      gender * bmi + activity * intensity, family = quasibinomial,
##      data = data)
##
## Coefficients:
##
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -6.954636   1.230179  -5.653 1.90e-08
```



```

## genderFemale          -0.026129    0.572923   -0.046    0.96363
## age                   0.111556    0.027361    4.077 4.82e-05
## ethnicOther Hispanic  -0.174447    0.287314   -0.607    0.54384
## ethnicNon-Hispanic White -0.259737    0.215972   -1.203    0.22932
## ethnicNon-Hispanic Black -0.712472    0.248972   -2.862    0.00428
## ethnicNon-Hispanic Asian  0.557777    0.243116    2.294    0.02192
## ethnicOther Race - Including Multi-Racial -0.294404    0.323717   -0.909    0.36327
## bmi                   0.136701    0.034420    3.971 7.51e-05
## activitymoderate      -1.026476    0.850220   -1.207    0.22752
## activityvigorous       0.657549    0.709871    0.926    0.35445
## intensitymoderate      2.006489    0.801847    2.502    0.01245
## intensityvigorous      0.860866    0.728758    1.181    0.23769
## age:bmi               -0.002364    0.000804   -2.940    0.00334
## age:activitymoderate   0.027218    0.017559    1.550    0.12135
## age:activityvigorous  -0.004231    0.015385   -0.275    0.78336
## age:intensitymoderate -0.050711    0.016816   -3.016    0.00261
## age:intensityvigorous -0.028825    0.016040   -1.797    0.07253
## genderFemale:bmi      -0.019395    0.017847   -1.087    0.27733
## activitymoderate:intensitymoderate  0.203016    0.463482    0.438    0.66144
## activityvigorous:intensitymoderate -0.115203    0.428591   -0.269    0.78813
## activitymoderate:intensityvigorous  0.155629    0.465880    0.334    0.73839
## activityvigorous:intensityvigorous -0.278204    0.406034   -0.685    0.49335
##
## (Intercept)          ***
## genderFemale
## age                  ***
## ethnicOther Hispanic
## ethnicNon-Hispanic White
## ethnicNon-Hispanic Black    **
## ethnicNon-Hispanic Asian    *
## ethnicOther Race - Including Multi-Racial
## bmi                  ***
## activitymoderate
## activityvigorous
## intensitymoderate        *
## intensityvigorous
## age:bmi                **
## age:activitymoderate
## age:activityvigorous
## age:intensitymoderate    **
## age:intensityvigorous    .
## genderFemale:bmi
## activitymoderate:intensitymoderate
## activityvigorous:intensitymoderate
## activitymoderate:intensityvigorous
## activityvigorous:intensityvigorous
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for quasibinomial family taken to be 0.9937757)
##
## Null deviance: 1342.0 on 1424 degrees of freedom
## Residual deviance: 1243.4 on 1402 degrees of freedom
## (1918 observations deleted due to missingness)

```

```
## AIC: NA
##
## Number of Fisher Scoring iterations: 5
```

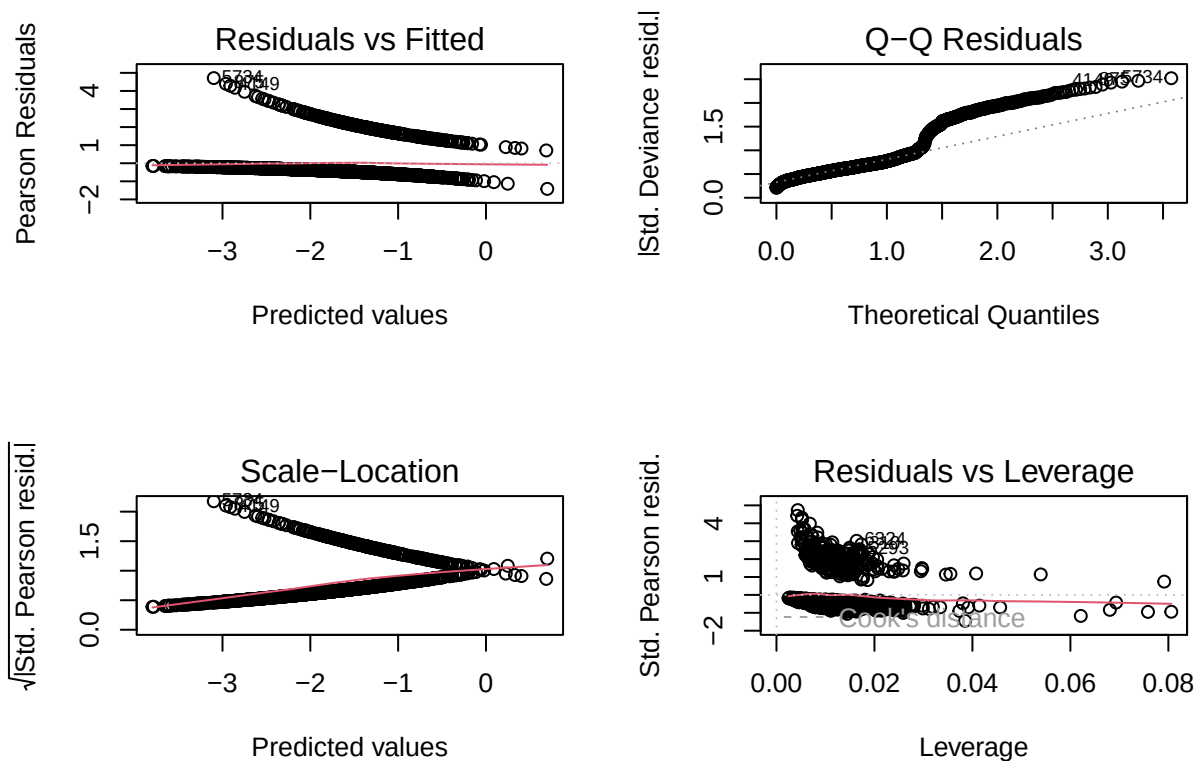
Age and activity interaction is not significant, neither is gender and bmi, nor activity and intensity so I remove these interaction terms

```
model1d <- glm(data=data,
               at_risk~gender+age+ethnic+bmi+activity+intensity+age*bmi+age*intensity,
               family=quasibinomial)
summary(model1d)
```

```
##
## Call:
## glm(formula = at_risk ~ gender + age + ethnic + bmi + activity +
##      intensity + age * bmi + age * intensity, family = quasibinomial,
##      data = data)
##
## Coefficients:
##
##              Estimate Std. Error t value
## (Intercept)      -6.7045718   1.1562635  -5.798
## genderFemale      -0.6325729   0.1496620  -4.227
## age               0.1140332   0.0265273   4.299
## ethnicOther Hispanic      -0.1634868   0.2854768  -0.573
## ethnicNon-Hispanic White  -0.2494768   0.2141884  -1.165
## ethnicNon-Hispanic Black  -0.7312518   0.2467033  -2.964
## ethnicNon-Hispanic Asian   0.5452741   0.2392566   2.279
## ethnicOther Race - Including Multi-Racial -0.2943969   0.3217552  -0.915
## bmi               0.1253745   0.0328000   3.822
## activitymoderate  0.2461979   0.1896443   1.298
## activityvigorous  0.3971426   0.1755113   2.263
## intensitymoderate 1.8084739   0.7288150   2.481
## intensityvigorous 0.6578641   0.6696518   0.982
## age:bmi          -0.0023605   0.0007872  -2.999
## age:intensitymoderate -0.0459618   0.0162961  -2.820
## age:intensityvigorous -0.0256211   0.0157228  -1.630
##
##              Pr(>|t|)
## (Intercept)      8.25e-09 ***
## genderFemale      2.52e-05 ***
## age              1.84e-05 ***
## ethnicOther Hispanic 0.566953
## ethnicNon-Hispanic White 0.244316
## ethnicNon-Hispanic Black 0.003087 **
## ethnicNon-Hispanic Asian 0.022814 *
## ethnicOther Race - Including Multi-Racial 0.360363
## bmi              0.000138 ***
## activitymoderate  0.194428
## activityvigorous  0.023801 *
## intensitymoderate 0.013203 *
## intensityvigorous 0.326073
## age:bmi          0.002759 **
## age:intensitymoderate 0.004863 **
## age:intensityvigorous 0.103419
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for quasibinomial family taken to be 0.9931544)
##
## Null deviance: 1342.0  on 1424  degrees of freedom
## Residual deviance: 1248.6  on 1409  degrees of freedom
## (1918 observations deleted due to missingness)
## AIC: NA
##
## Number of Fisher Scoring iterations: 5
```

```
par(mfrow = c(2, 2))
plot(model1d)
```



```
drop1(model1d)
```

```
## Single term deletions
##
## Model:
## at_risk ~ gender + age + ethnic + bmi + activity + intensity +
##          age * bmi + age * intensity
##           Df Deviance
## <none>          1248.6
## gender           1  1266.7
```

```
## ethnic          5    1276.6
## activity        2    1253.9
## age:bmi         1    1257.7
## age:intensity   2    1257.0
```

Deviance does not differ much with interaction terms so there is justification to keep them in the model. I am aware that the model plots are not ideal, however there is a lot of missing data which is difficult to capture in this model. The poor diagnostic plots may also be due to high-leverage points like the highest triglyceride data point that I mentioned previously. I will now explore how to deal with this missingness to improve this model.

Task 4

How do complete case analysis and multiple imputation compare in terms of the obtained model estimates for model selected in Task 3. [10 marks]

Complete case analysis and multiple imputation are ways of dealing with missing data. Complete case analysis involved removing all missing values from analysis. There is a loss of potentially useful information as a large proportion of observations are being dropped, reducing the sample size. AIC is not comparable, any changes in effect sizes and statistical significance may be due to loss of power rather than genuine.

```
# Select variables used in the model
completed_data <- data[, c("at_risk", "gender", "age", "ethnic", "bmi", "activity", "intensity")]
# Drop rows with any missing values in those columns
cca_data <- na.omit(completed_data)
model2a <- glm(data=completed_data,
               at_risk~gender+age+ethnic+bmi+activity+intensity+age*bmi+age*intensity,
               family=quasibinomial)
summary(model2a)
```

```
##
## Call:
## glm(formula = at_risk ~ gender + age + ethnic + bmi + activity +
##      intensity + age * bmi + age * intensity, family = quasibinomial,
##      data = completed_data)
##
## Coefficients:
##
##              Estimate Std. Error t value
## (Intercept)    -6.7045718   1.1562635  -5.798
## genderFemale    -0.6325729   0.1496620  -4.227
## age             0.1140332   0.0265273   4.299
## ethnicOther Hispanic    -0.1634868   0.2854768  -0.573
## ethnicNon-Hispanic White    -0.2494768   0.2141884  -1.165
## ethnicNon-Hispanic Black    -0.7312518   0.2467033  -2.964
## ethnicNon-Hispanic Asian     0.5452741   0.2392566   2.279
## ethnicOther Race - Including Multi-Racial -0.2943969   0.3217552  -0.915
## bmi            0.1253745   0.0328000   3.822
## activitymoderate    0.2461979   0.1896443   1.298
## activityvigorous    0.3971426   0.1755113   2.263
## intensitymoderate    1.8084739   0.7288150   2.481
## intensityvigorous    0.6578641   0.6696518   0.982
## age:bmi          -0.0023605   0.0007872  -2.999
## age:intensitymoderate -0.0459618   0.0162961  -2.820
```

```
## age:intensityvigorous      -0.0256211  0.0157228  -1.630
##                               Pr(>|t|)
## (Intercept)                8.25e-09 ***
## genderFemale               2.52e-05 ***
## age                        1.84e-05 ***
## ethnicOther Hispanic       0.566953
## ethnicNon-Hispanic White   0.244316
## ethnicNon-Hispanic Black    0.003087 **
## ethnicNon-Hispanic Asian    0.022814 *
## ethnicOther Race - Including Multi-Racial 0.360363
## bmi                        0.000138 ***
## activitymoderate           0.194428
## activityvigorous           0.023801 *
## intensitymoderate           0.013203 *
## intensityvigorous           0.326073
## age:bmi                    0.002759 **
## age:intensitymoderate       0.004863 **
## age:intensityvigorous       0.103419
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for quasibinomial family taken to be 0.9931544)
##
## Null deviance: 1342.0 on 1424 degrees of freedom
## Residual deviance: 1248.6 on 1409 degrees of freedom
## (1918 observations deleted due to missingness)
## AIC: NA
##
## Number of Fisher Scoring iterations: 5
```

Multiple imputation allows for uncertainty about missing data by creating several different plausible imputed data sets and appropriately combining results from each. It preserves the sample size, improving accuracy but is more complex to implement and I have to assume that the missing data is missing at random.

```
imp_vars <- data[, c("at_risk", "gender", "age", "ethnic", "bmi", "activity", "intensity")]

# Convert categorical variables to factors
imp_vars$gender <- as.factor(imp_vars$gender)
imp_vars$ethnic <- as.factor(imp_vars$ethnic)
imp_vars$activity <- as.factor(imp_vars$activity)
imp_vars$intensity <- as.factor(imp_vars$intensity)
imp_vars$bmi <- as.numeric(imp_vars$bmi)

# Perform multiple imputation (m = number of imputed datasets)
imputation_data <- mice(imp_vars, m = 5, method = "pmm", print=F)
model2b <- with(imputation_data,
               glm(at_risk~gender+age+ethnic+bmi+activity+intensity+age*bmi+age*intensity,
                  family = quasibinomial()))
pool.fit <- pool(model2b)
summary(pool.fit)
```

```
##                               term      estimate  std.error
## 1                               (Intercept) -5.217158534  0.8243550835
```

```
## 2          genderFemale -0.507016944 0.1512718475
## 3          age 0.074379735 0.0180200403
## 4          ethnicOther Hispanic -0.233910100 0.3638768279
## 5          ethnicNon-Hispanic White -0.236579070 0.2337433020
## 6          ethnicNon-Hispanic Black -0.472544333 0.1862316869
## 7          ethnicNon-Hispanic Asian 0.629691922 0.2524627109
## 8 ethnicOther Race - Including Multi-Racial -0.184244968 0.2805263652
## 9          bmi 0.089066871 0.0266854218
## 10         activitymoderate 0.272104412 0.1970948593
## 11         activityvigorous 0.476631144 0.2134860327
## 12         intensitymoderate 0.568310516 0.6705190915
## 13         intensityvigorous 0.087095533 0.4768602631
## 14         age:bmi -0.001452653 0.0006058386
## 15         age:intensitymoderate -0.019836172 0.0126022324
## 16         age:intensityvigorous -0.011826458 0.0111024007
##      statistic      df      p.value
## 1 -6.3287758 122.549650 4.221676e-09
## 2 -3.3516940 11.217563 6.292695e-03
## 3 4.1276120 399.497835 4.464078e-05
## 4 -0.6428277 7.972991 5.383788e-01
## 5 -1.0121320 10.376136 3.344947e-01
## 6 -2.5374003 50.584620 1.429446e-02
## 7 2.4941977 10.549251 3.064087e-02
## 8 -0.6567831 29.882056 5.163471e-01
## 9 3.3376602 33.588440 2.075069e-03
## 10 1.3805759 10.549925 1.959565e-01
## 11 2.2326104 7.707534 5.730352e-02
## 12 0.8475680 19.473361 4.069757e-01
## 13 0.1826437 85.897449 8.555080e-01
## 14 -2.3977566 48.940630 2.035574e-02
## 15 -1.5740205 74.334746 1.197296e-01
## 16 -1.0652163 139.725303 2.886153e-01
```

In this pool fit, it is worth noting how intensity is now not significant as well as the interaction terms. I will extract estimates from both models and include them in one table for easy comparison:

```
estimates_completed <- model2a$coefficients
estimates_imputation <- summary(pool.fit)$estimate
estimates_table <- data.frame(
  Complete_Case = estimates_completed,
  Multiple_Imputation = estimates_imputation
)
kable(estimates_table, digits = 3, caption = "Comparison of Model Estimates")
```

Table 1: Comparison of Model Estimates

	Complete_Case	Multiple_Imputation
(Intercept)	-6.705	-5.217
genderFemale	-0.633	-0.507
age	0.114	0.074
ethnicOther Hispanic	-0.163	-0.234
ethnicNon-Hispanic White	-0.249	-0.237

	Complete_Case	Multiple_Imputation
ethnicNon-Hispanic Black	-0.731	-0.473
ethnicNon-Hispanic Asian	0.545	0.630
ethnicOther Race - Including Multi-Racial	-0.294	-0.184
bmi	0.125	0.089
activitymoderate	0.246	0.272
activityvigorous	0.397	0.477
intensitymoderate	1.808	0.568
intensityvigorous	0.658	0.087
age:bmi	-0.002	-0.001
age:intensitymoderate	-0.046	-0.020
age:intensityvigorous	-0.026	-0.012

Task 5

Write a non-technical summary of your findings from Task 3 and 4. [5 marks]

Task 3: During my investigation into the risk of cardiovascular disease, I analysed data from US National Health and Nutrition Examination Survey. I created a variable to signify whether a participant was at risk based on their cholesterol and triglyceride levels. I then explored what variables may influence someone's risk to cardiovascular disease and found gender, age, ethnicity, bmi, activity and intensity levels all to be significant influences on a person's risk. The results from this exploration found that:

- Females are less at risk than males
- As age increases the risk of cardiovascular disease also increases
- Non-Hispanic Asians have an increased risk compared to Mexican Americans, the other ethnicities have a lower risk than Mexican Americans
- The higher a person's bmi, the more at risk of cardiovascular disease they are
- Moderate activity makes a person less at risk, and vigorous activity more at risk of cardiovascular disease
- Intensity of activity makes a person more at risk, with moderate activity having the bigger effect on risk of cardiovascular disease

Task 4: However, in the data there were a lot of missing values that were affecting the results- I tried two different methods to overcome this. One of these was by simply ignoring participants who had any missing values for the variables of interest. This causes a loss of important data, and resulted in a smaller sample size. Another approach I tried was by filling in missing values based on their other values that were not missing- therefore the sample size remains the same, which is preferred. However, there is concern over uncertainty with the estimates due to using predicted values and the complexity of this approach. I did find that the model gave similar estimates to a participant being at risk for both approaches which is good and effects remain the same, i.e. all positive effects on cardiovascular risk remain positive.